Population mean μ= 3.25 cm, Population SD σ = 2.61 cm.

Null Hypothesis $H_0 : \mu$ = 3.25 cm (the sample has been drawn from the population mean μ = 3.25 cm and SD σ = 2.61 cm)

Alternative Hypothesis $H_1 : \mu \neq 3.25$ cm (two tail) i.e., the sample has not been drawn from the population mean μ = 3.25 cm and SD σ = 2.61 cm.

The level of significance $\alpha = 5\% = 0.05$

Teststatistic:

$$Z = \frac{3.4 - 3.25}{\frac{2.61}{\sqrt{900}}} = \frac{0.15}{0.087} = 1.724$$

$$\therefore Z = 1.724$$

Thus the calculated and the significant value or table value $Z_{\alpha/2} = 1.96$

Comparing the calculated and table values, $Z < Z_{\alpha/2}$ i.e., 1.724 < 1.96

Inference: Since the calculated value is less than table value i.e., $Z < Z_{\alpha/2}$ at 5% level of significance, the null hypothesis is accepted. Hence we conclude that the data doesn't provide us any evidence against the null hypothesis. Therefore, the sample has been drawn from the population mean μ = 3.25 cm and SD, σ = 2.61 cm.

(ii) Confidence limits

95% confidential limits for the population mean μ are :

$$\bar{x} - Z_{\alpha/2} SE \leq \mu \leq \bar{x} + Z_{\alpha/2} SE$$

$$3.4 - (1.96 \times 0.087) \leq \mu \leq 3.4 + (1.96 \times 0.087)$$

$$3.229 \leq \mu \leq 3.571$$

98% confidential limits for the population mean μ are :

$$\bar{x} - Z_{\alpha/2} SE \leq \mu \leq \bar{x} + Z_{\alpha/2} SE$$

$$3.4 - (2.33 \times 0.087) \leq \mu \leq 3.4 + (2.33 \times 0.087)$$

$$3.197 \leq \mu \leq 3.603$$

Therefore, 95% confidential limits is (3.229,3.571) and 98% confidential limits is (3.197, 3.603).

Example 8.17

The mean weekly sales of soap bars in departmental stores were 146.3 bars per store. After an advertising campaign the mean weekly sales in 400 stores for a typical week increased to 153.7 and showed a standard deviation of 17.2. Was the advertising campaign successful at 95% confidence limit?

Solution:

Sample size	n	= 400 stores
Sample mean	$\bar{x}$	= 153.7 bars
Sample SD	s	= 17.2 bars
Population mean	m	= 146.3 bars

Since population *SD* is unknown we can consider the sample *SD* $s = \sigma$

Null Hypothesis. The advertising campaign is not successful i.e, $H_0 : \mu$ = 146.3 (There is no significant difference between the mean weekly sales of soap bars in department stores before and after advertising campaign)

Alternative Hypothesis $H_1 : \mu$ > 143.3 (Right tail test). The advertising campaign was successful

Level of significance α = 0.05

Test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

$$Z = \frac{153.7 - 146.3}{\frac{17.2}{\sqrt{400}}}$$

$$= \frac{7.4}{0.86} = 8.605$$

$$\therefore Z = 8.605$$

Comparing the calculated value $Z = 8.605$ and the significant value or table value $Z_\alpha = 1.645$. we get 8.605 > 1.645.

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